

Theorem C.2 as a self-contained problem

Problem. For $n \geq 0$, let $\text{NC}(n)$ be the set of noncrossing partitions of $\{1, \dots, n\}$, i.e. partitions with no $a < b < c < d$ such that a, c lie in one block and b, d lie in another. For a block B , put

$$\delta(B) = 2 \left\lfloor \frac{|B| - 1}{2} \right\rfloor, \quad \delta(\pi) = \sum_{B \in \pi} \delta(B).$$

For indeterminates λ, t , define

$$M_n(\lambda, t) = \sum_{\pi \in \text{NC}(n)} \lambda^{|\pi|} t^{\delta(\pi)/2} \in \mathbb{Q}[\lambda, t].$$

For $m \geq 0$, set

$$H_m(\lambda, t) = (M_{i+j+1}(\lambda, t))_{0 \leq i, j \leq m}.$$

Define $f_{0,m}, f_{1,m} \in \mathbb{Q}[\lambda]$ by

$$f_{0,m}(\lambda) = [t^0] \det H_m(\lambda, t), \quad f_{1,m}(\lambda) = [t^1] \det H_m(\lambda, t).$$

Finally define $d_r(\lambda)$ by

$$d_0(\lambda) = 1, \quad d_1(\lambda) = \lambda, \quad d_{r+2}(\lambda) = \lambda d_{r+1}(\lambda) - \lambda d_r(\lambda),$$

and define $g_m(\lambda) \in \mathbb{Q}[\lambda]$ by

$$d_{m+1}(\lambda) = \lambda^{\lceil (m+1)/2 \rceil} g_m(\lambda), \quad g_m(0) \neq 0.$$

Theorem C.2. For every $m \geq 0$,

$$g_m(\lambda) \mid (f_{1,m}(\lambda) - f'_{0,m}(\lambda)) \quad \text{in } \mathbb{Q}[\lambda].$$

Here t is the bookkeeping variable for the first finite-size defect in Appendix C. Thus $M_n(\lambda, 0)$ is the balanced moment, $f_{0,m}$ is the balanced Hankel determinant, $f_{1,m}$ is its first finite-size correction, and g_m is the threshold factor obtained from the tridiagonal determinant d_{m+1} . With these identifications, the displayed divisibility is exactly Theorem C.2.